

Complete System Bill of Materials

All prices verified against vendor listings and actual purchases, June 2026. Raspberry Pi and all SSD/NAND storage reflect the current industry memory shortage (SSDs up 2-4× from 2025) and are expected to decline in 2027-2028. Icom frequently runs instant coupons — check hamradio.com/special_offers before purchase.

ITEM	DETAIL	QTY	UNIT	EXT
FieldCommand Server (192.168.50.1)				
Raspberry Pi 5 — 16 GB	Model B, 16 GB RAM — main application server	1	\$305	\$305
Pironman 5-MAX tower	SunFounder — dual NVMe RAID 0/1, OLED, tower cooler, RGB — includes a battery-backed real-time clock (RTC) so the server keeps time across power-offs	1	\$95	\$95
1 TB NVMe SSD	2× M.2 2280 PCIe — RAID 1 mirror (NAND shortage pricing — was ~\$75 in 2025)	2	\$180	\$360
Pi 27W USB-C PSU	Official Raspberry Pi 27W	1	\$14	\$14
MicroSD 32 GB	Boot media / recovery	1	\$10	\$10
	FieldCommand Server (192.168.50.1) subtotal			\$784
44Net Gateway (192.168.50.2)				
Raspberry Pi 5 — 16 GB	Dedicated AMPRNet WireGuard gateway (8 GB variant ~\$130 acceptable)	1	\$305	\$305
Argon NEO 5 case	With M.2 SATA expansion	1	\$35	\$35
256 GB M.2 SATA SSD	Gateway OS drive (NAND shortage pricing)	1	\$45	\$45
Pi 27W USB-C PSU	Official Raspberry Pi 27W	1	\$14	\$14
	44Net Gateway (192.168.50.2) subtotal			\$399
Networking				
ASUS RT-BE58 Go	Wi-Fi 7 travel router — 1 primary + 2 AiMesh nodes (street \$119; MSRP \$160)	3	\$119	\$357
UniFi Switch Lite 16 PoE	16-port GbE, 8× PoE — central wired hub	1	\$199	\$199
CAT6 cables	Assorted lengths	10	\$3	\$30
	Networking subtotal			\$586
WAN — Cellular + Satellite				
InstyConnect Drum	Omnidirectional 5G/LTE antenna+modem — dual-carrier cellular — primary WAN	1	\$375	\$375
InstyConnect Switchblade	Directional folding antenna — backup for weak-signal sites	1	\$425	\$425
Spare PoE cable	Outdoor-rated shielded CAT6	1	\$20	\$20
Starlink Standard dish kit	Standard dish — automatic failover WAN (price DROP from \$499; regional promos lower)	1	\$349	\$349
Starlink Ethernet adapter	Official adapter	1	\$25	\$25
USB-to-Ethernet adapter	GbE USB 3.0 — ASUS USB WAN port	1	\$18	\$18
	WAN — Cellular + Satellite subtotal			\$1,212
Operator Workstations				
Raspberry Pi 500+	Keyboard PC — 16 GB RAM, 256 GB NVMe, mechanical keyboard (Gateron KS-33 Blue)	4	\$410	\$1,640
Raspberry Pi 500+ Desktop Kit upgrade	Adds mouse, 27W PSU, HDMI cable per unit (buy as kit @ \$430 ea)	4	\$20	\$80
Raspberry Pi Monitor	15.6" 1080p IPS, USB-C powered	4	\$100	\$400
	Operator Workstations subtotal			\$2,120

ITEM	DETAIL	QTY	UNIT	EXT
HF Station				
Icom IC-7300MK2	HF/50 MHz 100 W SDR — HDMI out, USB-C, LAN remote, CW decoder, improved RMDR — check Icom coupons at hamradio.com/special_offers	1	\$1,500	\$1,500
Astron RS-35M-AP power supply	13.8 VDC linear regulated, 25 A cont / 35 A peak, dual meters, Anderson Powerpoles — qty 2: primary for IC-7300MK2 + redundant/spare for field ops	2	\$344	\$688
LDG IT-100	125 W Icom autotuner — station-side, powered/controlled by transceiver	1	\$180	\$180
USB cable	Radio to laptop, shielded	1	\$10	\$10
Windows laptop	Winlink Express + VARA HF + JS8Call — budget \$1,000 for capable unit with SSD and USB ports	1	\$1,000	\$1,000
	HF Station subtotal			\$3,378
PACTOR HF Modem				
SCS PXdragon DR-9400	Current-generation PACTOR-4 HF modem — replaces discontinued DR-7800/7400 series — PACTOR-1/-2/-3/-4, Packet Radio 1200/9600 baud, Robust Packet Radio — quad-core ARM v8 @ 1.8 GHz real-time Linux — 1-ppm TCXO — USB-C + Ethernet LAN standard — optional Wi-Fi/Bluetooth dongle — OLED display — Winlink, Airmail, RMS Express compatible — manufactured in Germany by SCS — US dealer warranty via SeaTech Systems (no overseas shipping)	1	\$2,099	\$2,099
IC-7300MK2 interface cable set	Audio cable (mic/speaker) + CI-V control cable from modem DIN port to IC-7300MK2 — with ferrite filters — ordered from SeaTech Systems or DX Engineering at time of purchase	1	\$60	\$60
	PACTOR HF Modem subtotal			\$2,159
VHF / UHF Station				
Yaesu FTM-510DR	144/430 MHz dual band C4FM digital + FM — 55W VHF / 50W UHF — built-in GPS + APRS (1200/9600 baud) — Super-DX weak-signal mode — 2.4" color touch display — detachable panel — WiRES-X / Fusion digital — in-stock HRO (FTM-510DR ASP w/ DSP board \$749.95 if preferred)	1	\$670	\$670
Comet CA-2X4SRNMO antenna	Wideband 2m/70cm mobile vertical designed for SAR/EMCOMM — 140-160 / 435-465 MHz covers Starcom (155.xxx MHz) AND ham bands — 3.8/6.2 dBi — 150W — 40" fold-over — NMO connector	1	\$72	\$72
Comet CM-5NMO mag mount	3.25" heavy-duty magnetic base, NMO connector, RG-58 coax tail — vehicle roof or equipment case lid mount	1	\$35	\$35
RG-8X coax jumper	Short jumper: CM-5NMO coax tail to FTM-510DR SO-239 antenna port	1	\$15	\$15
Digirig Mobile	Rev 1.11 — USB digital modes interface — combines audio codec, serial CAT and PTT — single USB-C connection — works with Windows/Mac/Linux — enables FT8, JS8Call, SSTV, VARA FM, packet on VHF/UHF via the FTM-510DR	1	\$50	\$50
Mobilinkd TNC4	1200/9600 baud KISS TNC — Bluetooth 4.2 (iOS + Android) + USB-C — AFSK/GFSK/4-FSK — DSP modem — TCXO precision timing — 900mAh battery (48 hr) — enables APRS KISS mode for Direwolf / APRS Command on FieldCommand	1	\$150	\$150
Digirig Yaesu FTM MiniDin10 cable	digirig.net SKU YAESU10MOBILE — 10-pin MiniDin to Digirig audio socket — explicitly listed for FTM-510DR — audio + PTT — enables VARA FM, Winlink, JS8Call, SSTV, APRS via laptop	1	\$20	\$20
Mobilinkd Yaesu FTM modem cable	store.mobilinkd.com — 10-pin MiniDIN, 1.5m — fits FTM-510DR (same connector as FTM-100/200/300/400/500 series) — connects TNC4 to radio for APRS KISS mode	1	\$25	\$25
USB-C data cable for Digirig	Standard USB-C cable supporting both power and data — NOT included with Digirig — required to connect Digirig to laptop	1	\$10	\$10
	VHF / UHF Station subtotal			\$1,027

ITEM	DETAIL	QTY	UNIT	EXT
Transport Cases				
Harbor Freight Apache 4800 XL	Weatherproof protective case — IP65 rated — interior 17-7/8 x 12-7/8 x 6-5/8 in — pick-and-pull foam insert — purge valve — latching clasps — polypropylene shell — comparable to Pelican 1550 at a fraction of the cost — available in black, tan, yellow, orange, green — regular \$59.99; coupons bring to \$49.99 or \$39.99 (Inside Track Club)	6	\$60	\$360
	Transport Cases subtotal			\$360
Field Antenna System				
Chameleon CHA MPAS 2.0	Modular Portable Antenna System — 1.8-54 MHz — vertical/sloper/inverted-V/NVIS/man-pack — deploys under 2 min — hybrid match, MIL 2.0 + EXT 2.0 whips, spike mount, wire, coax, bag (USA250 25%-off sale thru mid-July 2026)	1	\$660	\$660
Chameleon CHA URT1	Universal Remote Tuner — weatherproof feedpoint tuner — 5-1,500 Ω , 125 W, 16,000 memories — power/control over coax	1	\$390	\$390
Chameleon antenna wire	300 ft tinned-copper Kevlar PTFE — long wires, NVIS, dipoles, radials	1	\$300	\$300
Wire winders	Heavy-duty, 75-100 ft capacity each	4	\$15	\$60
	Field Antenna System subtotal			\$1,410
Printing & Accessories				
Brother MFC-L3780CDW	Color laser MFP — IAP packages, ICS forms, access cards (L3770 discontinued; successor model)	1	\$450	\$450
LaCie Rugged 1 TB USB-C	LaCie Rugged USB-C 1TB (STFR1000800) — drop resistant 4 ft, crush 1 ton, rain resistant — 130 MB/s — USB-C + USB-A adapter included — Seagate Rescue data recovery service included — ideal for field deployment backup	1	\$125	\$125
USB flash 32 GB	Sneakernet transfer	1	\$10	\$10
USB GPS receiver	u-blox USB GPS — the server's offline time source: supplies exact UTC to chrony with no internet, plus position for the dashboard. Place the antenna with a clear view of the sky (window, outdoors, or rooftop) so it can get a fix.	1	\$35	\$35
	Printing & Accessories subtotal			\$590
Core infrastructure (server, gateway, network, WAN, workstations, printing/accessories)				\$5,691
HF station (IC-7300MK2, 2× Astron RS-35M-AP, LDG IT-100, cable, laptop)				\$3,378
VHF/UHF station (FTM-510DR, Comet CA-2X4SRNMO, mag mount, coax)				\$792
Transport cases (6× Harbor Freight Apache 4800 XL @ \$59.99 list)				\$360
Field antenna system (MPAS 2.0, URT1, wire, winders)				\$1,410
COMPLETE SYSTEM TOTAL				\$14,075

Recurring costs: the InstyConnect data plan — \$5/month standby, \$79-99/month during activations only · the Starlink service — ~\$120/month or pausable plan · No software licenses, subscriptions, or per-seat fees.

Pricing sources: raspberrypi.com · R&L; Electronics · DX Engineering · GigaParts · Ham Radio Outlet · chameleonantenna.com · instyconnect.com · starlink.com — verified June 2026.